LEMMA ML COUNTABLE IN ALGEBRA.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter algebra.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the extractor's stacks-candidates line, not from the earlier judged initial_corpus line; the H2 judgement and its reason below were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:algebra.tex:lemma-ML-countable.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 059W: sources/algebra.tex, source lines 22053–22053. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 0596: sources/algebra.tex, source lines 20736–20736. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0597: sources/algebra.tex, source lines 20748–20748. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let R be a ring. Let M be an R-module. Assume M is Mittag-Leffler and countably generated. For any R-module map $f : P \to M$ with P finitely generated there exists an endomorphism $\alpha : M \to M$ such that*

(1) *$\alpha : M \to M$ factors through a finitely presented R-module, and*

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(2) $\alpha \circ f = f$.

Proof. Write $M = \operatorname{colim}_{i \in I} M_i$ as a directed colimit of finitely presented R-modules with I countable, see Lemma 059W. The transition maps are denoted f_{ij} and we use $f_i : M_i \to M$ to denote the canonical maps into M. Set $N = \prod_{s \in I} M_s$. Denote

$$M_i^* = \operatorname{Hom}_R(M_i, N) = \prod\nolimits_{s \in I} \operatorname{Hom}_R(M_i, M_s)$$

so that (M_i^*) is an inverse system of R-modules over I. Note that $\operatorname{Hom}_R(M, N) = \lim M_i^*$. As M is Mittag-Leffler, we find for every $i \in I$ an index $k(i) \geq i$ such that

$$E_i := \bigcap\nolimits_{i' \geq i} \operatorname{Im}(M_{i'}^* \to M_i^*) = \operatorname{Im}(M_{k(i)}^* \to M_i^*)$$

Choose and fix $j \in I$ such that $\operatorname{Im}(P \to M) \subset \operatorname{Im}(M_j \to M)$. This is possible as P is finitely generated. Set $k = k(j)$. Let $x = (0, \ldots, 0, \operatorname{id}_{M_k}, 0, \ldots, 0) \in M_k^*$ and note that this maps to $y = (0, \ldots, 0, f_{jk}, 0, \ldots, 0) \in M_j^*$. By our choice of k we see that $y \in E_j$. By Example 0596 the transition maps $E_i \to E_j$ are surjective for each $i \geq j$ and $\lim E_i = \lim M_i^* = \operatorname{Hom}_R(M, N)$. Hence Lemma 0597 guarantees there exists an element $z \in \operatorname{Hom}_R(M, N)$ which maps to y in $E_j \subset M_j^*$. Let z_k be the kth component of z. Then $z_k : M \to M_k$ is a homomorphism such that

$$\begin{array}{ccc} M & \xrightarrow{z_k} & M_k \\ {\scriptstyle f_j}\uparrow & \nearrow_{f_{jk}} & \\ M_j & & \end{array}$$

commutes. Let $\alpha : M \to M$ be the composition $f_k \circ z_k : M \to M_k \to M$. Then α factors through a finitely presented module by construction and $\alpha \circ f_j = f_j$. Since the image of f is contained in the image of f_j this also implies that $\alpha \circ f = f$. $\square$